

CSE 318: Artificial Intelligence Sessional

Offline 1 Report: Performance Evaluation of Weighted A^* Search using the Custom Heuristic on the 15-Puzzle

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1 Introduction

This report evaluates a custom heuristic design, **CornerLinearConflict**, implemented within a Weighted A^* search framework to solve the 15-Puzzle (4×4 sliding tile puzzle). Performance metrics are analyzed across ten distinct initial board configurations using four expansion weight parameters: $W \in \{1.0, 1.2, 2.0, 5.0\}$.

2 Heuristic Formulation

The evaluated heuristic function, denoted as $h_{CLC}(n)$, is defined mathematically as follows:

$$h_{CLC}(n) = h_M(n) + 2 \cdot (C_{LC}(n) + C_{corner}(n)) \quad (1)$$

Where:

- $h_M(n)$ is the standard Manhattan Distance baseline.
- $C_{LC}(n)$ represents the count of classical Linear Conflicts in the rows and columns.
- $C_{corner}(n)$ is the count of linear conflicts occurring strictly between tiles whose target goal destinations are one of the four grid corners.

3 Initial Board Configurations

Number of test cases: 10 randomly generated solvable configurations ¹

$$\begin{array}{lll} \text{Case 1} = \begin{pmatrix} 0 & 8 & 5 & 12 \\ 2 & 14 & 3 & 1 \\ 9 & 4 & 11 & 15 \\ 7 & 10 & 6 & 13 \end{pmatrix} & \text{Case 2} = \begin{pmatrix} 4 & 11 & 8 & 2 \\ 0 & 9 & 12 & 15 \\ 7 & 5 & 1 & 13 \\ 6 & 3 & 10 & 14 \end{pmatrix} & \text{Case 3} = \begin{pmatrix} 2 & 4 & 6 & 8 \\ 14 & 3 & 13 & 5 \\ 7 & 9 & 11 & 0 \\ 12 & 15 & 10 & 1 \end{pmatrix} \\ \text{Case 4} = \begin{pmatrix} 4 & 12 & 3 & 7 \\ 11 & 0 & 1 & 5 \\ 8 & 13 & 2 & 6 \\ 10 & 15 & 9 & 14 \end{pmatrix} & \text{Case 5} = \begin{pmatrix} 4 & 8 & 10 & 13 \\ 11 & 6 & 1 & 9 \\ 7 & 12 & 15 & 2 \\ 5 & 0 & 14 & 3 \end{pmatrix} & \text{Case 6} = \begin{pmatrix} 9 & 3 & 15 & 1 \\ 10 & 14 & 0 & 4 \\ 13 & 2 & 7 & 5 \\ 6 & 11 & 8 & 12 \end{pmatrix} \\ \text{Case 7} = \begin{pmatrix} 1 & 13 & 9 & 0 \\ 10 & 5 & 11 & 4 \\ 12 & 2 & 6 & 3 \\ 15 & 14 & 7 & 8 \end{pmatrix} & \text{Case 8} = \begin{pmatrix} 1 & 8 & 6 & 15 \\ 4 & 12 & 14 & 0 \\ 3 & 7 & 13 & 5 \\ 10 & 9 & 2 & 11 \end{pmatrix} & \text{Case 9} = \begin{pmatrix} 14 & 10 & 7 & 2 \\ 0 & 5 & 6 & 9 \\ 4 & 1 & 12 & 15 \\ 3 & 8 & 11 & 13 \end{pmatrix} \\ & \text{Case 10} = \begin{pmatrix} 10 & 13 & 2 & 15 \\ 9 & 1 & 5 & 7 \\ 14 & 4 & 12 & 3 \\ 6 & 11 & 0 & 8 \end{pmatrix} \end{array}$$

¹The puzzle configurations were generated using <https://15puzzle.netlify.app/>. The board arrays were extracted from screenshots with assistance from Google's Gemini.

4 Empirical Search Metrics

Table 1 outlines the behavior of the search space across varying weight bounds.

Table 1: Weighted A^* Search Metrics across Configurations

Configuration	Weight (W)	Moves	Nodes Explored	Runtime (ms)
Case 1	1.0	48	143,357	2,717.0
	1.2	48	143,357	2,945.1
	2.0	50	495	7.9
	5.0	110	562	12.9
Case 2	1.0	49	74,885	1,388.7
	1.2	49	74,885	1,426.4
	2.0	61	1,192	36.9
	5.0	93	463	8.3
Case 3	1.0	53	2,425,638	50,588.8
	1.2	53	2,425,638	59,384.5
	2.0	61	904	17.2
	5.0	93	362	8.2
Case 4	1.0	50	35,498	700.3
	1.2	50	35,498	629.9
	2.0	66	1,105	21.8
	5.0	102	1,070	21.4
Case 5	1.0	58	6,473,694	140,571.0
	1.2	58	6,473,694	167,406.8
	2.0	70	7,420	142.0
	5.0	94	365	6.9
Case 6	1.0	45	107,906	2,483.5
	1.2	45	107,906	2,316.8
	2.0	51	792	14.4
	5.0	51	86	1.9
Case 7	1.0	47	245,337	5,538.8
	1.2	47	245,337	5,453.1
	2.0	51	1,314	27.0
	5.0	73	401	9.2
Case 8	1.0	50	110,953	2,403.6
	1.2	50	110,953	2,794.2
	2.0	68	10,351	223.7
	5.0	92	1,816	39.5
Case 9	1.0	55	1,606,130	35,473.9
	1.2	55	1,606,130	39,797.1
	2.0	65	2,136	39.9
	5.0	93	363	6.3
Case 10	1.0	47	375,353	8,145.8
	1.2	47	375,353	8,408.3
	2.0	61	1,155	22.7
	5.0	75	320	6.8

5 Experimental Visualizations and Analysis

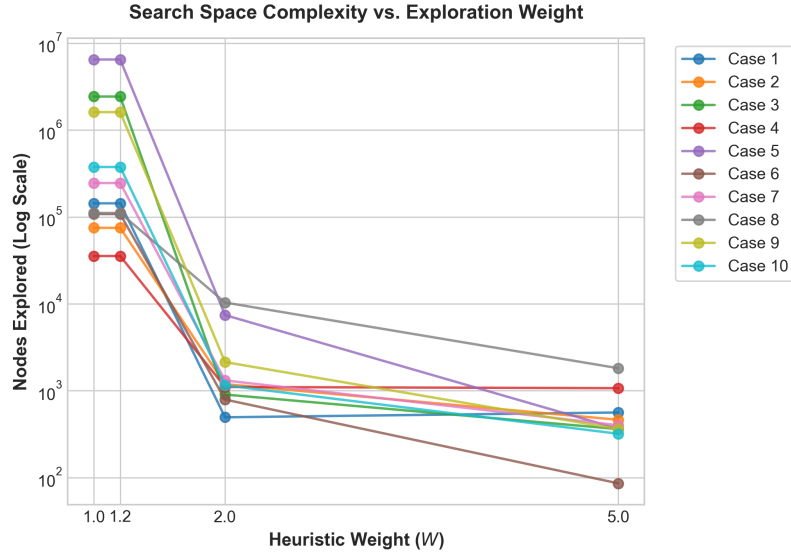


Figure 1: Total Nodes Explored across configurations plotted against Heuristic Weight (W) using a logarithmic scale.

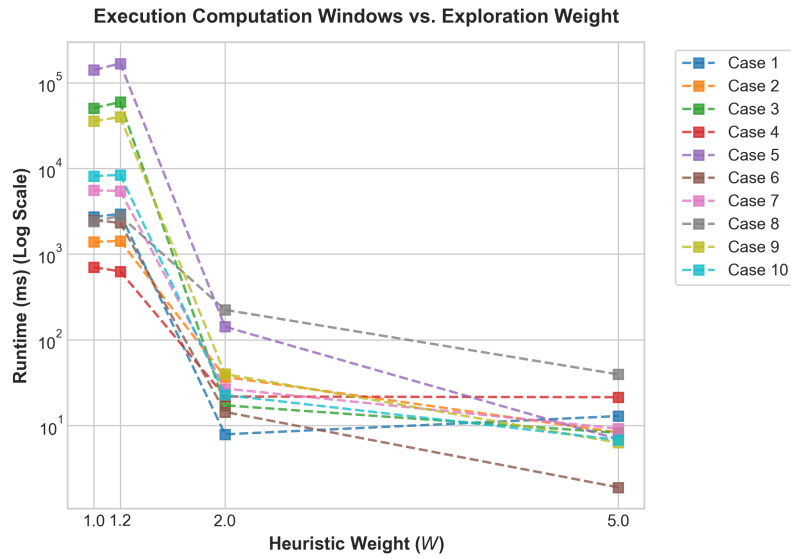


Figure 2: Algorithm Execution Runtime (ms) across configurations plotted against Heuristic Weight (W) using a logarithmic scale.

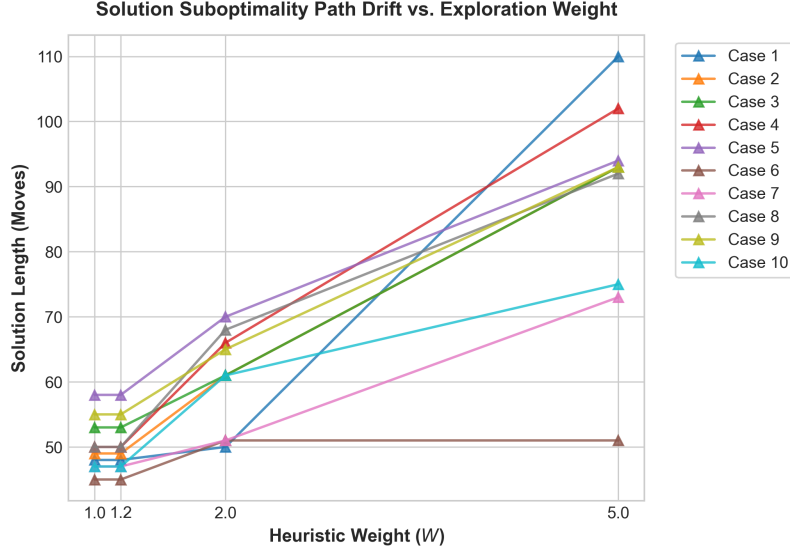


Figure 3: Total moves needed to solve puzzles across configurations plotted against Heuristic Weight (W).

5.1 Key Performance Trends

1. **Heuristic Scale Insensitivity ($W = 1.0$ vs $W = 1.2$):** Across all test scenarios, the node expansion counts match exactly between $W = 1.0$ and $W = 1.2$. This occurs because the integer values generated by our inflated heuristic dominate the search evaluation equation $f(n) = g(n) + W \cdot h(n)$. Scaling $h(n)$ by a factor of 1.2 does not alter the relative processing sequence within the priority queue.
2. **Exponential Decay of Search Volume ($W \geq 2.0$):** As W escalates, the search strategy shifts towards a greedy best-first pattern. In Case 5, increasing the weight boundary from 1.0 to 5.0 reduces the explored state footprint from 6,473,694 to a mere 365 nodes—representing an execution efficiency return of over 99.99%.
3. **Suboptimality Inflation:** The performance savings achieved by setting $W = 5.0$ introduce severe suboptimality. For instance, Case 1 expands from an optimal path of 48 moves to a winding solution of 110 moves.